AlphaDesk

How the internal models work: data, training, combination and decision. Not the AI agents.

This document describes every forecasting and ranking model that AlphaDesk computes itself, on your machine, from market data. The AI agents (the LLM analysts on a ticker's page) are left out: they are shown for comparison and decide nothing. Numbers come from the code on branch *portfolio-app* and from the saved studies in *experiments/output* (29 October 2026).

Read this first. None of these models is validated for trading. Their tested skill is small (rank correlations of about 0.02-0.05 over 2012-24) and most of it fades in 2025-26. The daily decision therefore runs in *experimental* mode: it shows PAPER ENTER instead of ENTER, and a paper account follows it to collect the evidence that could validate it.

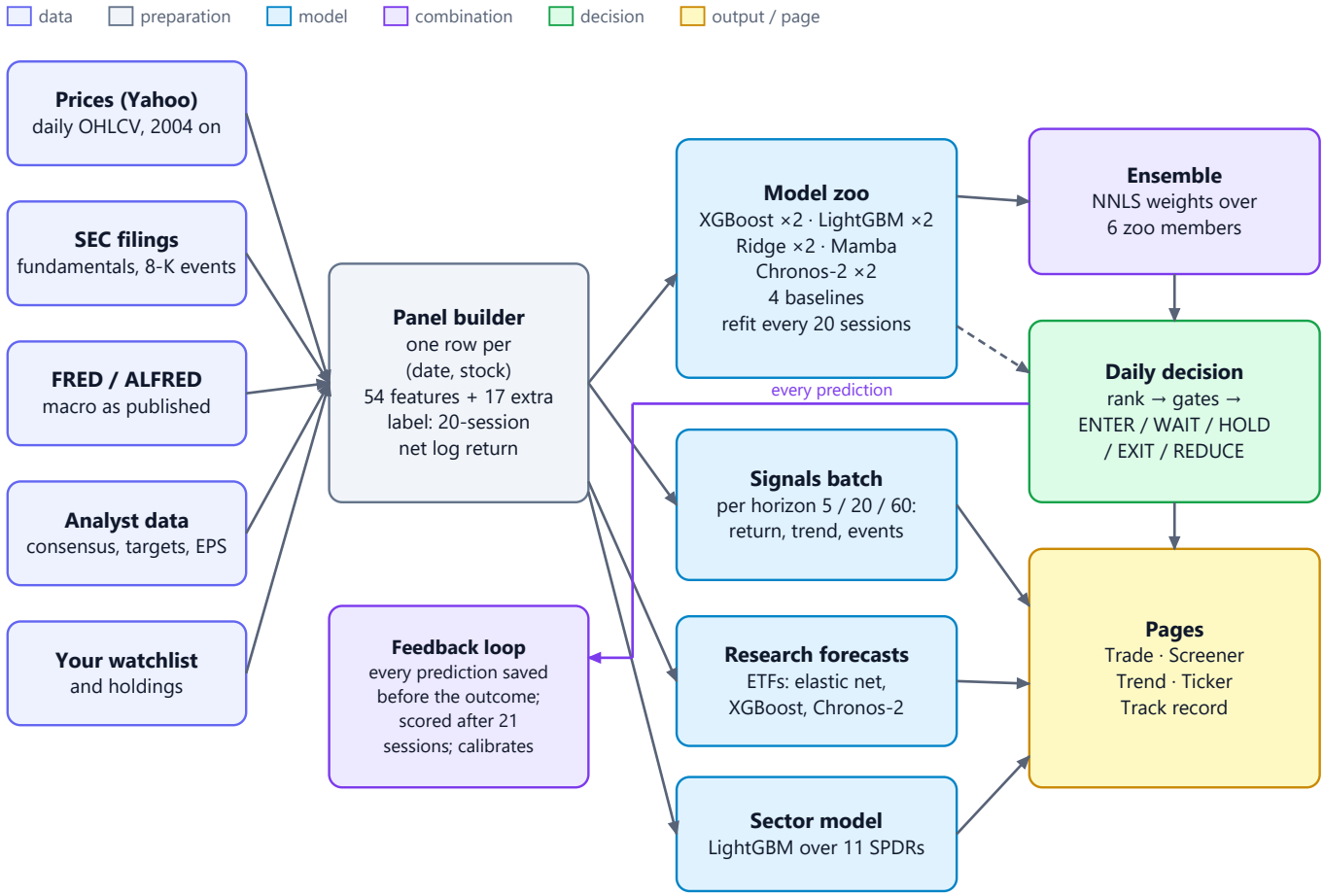


Figure 1. The whole system: five data sources feed one panel. Four model families read it; the zoo's members are combined into the ensemble, which orders the daily decision (dashed: the fallback when the ensemble cannot score).

1. The data and the panel

Every model reads the same **panel**: one row per decision date and stock, with only the information available at that date's close. Rows need a price that day and 252 sessions of history. Stocks are the point-in-time S&P 500 members when the membership history is available (survivorship partly corrected), plus your watchlist, scored but never trained on.

Group	Count	What it holds	Source
Technical	26	returns over 1-12 months, volatility, trend, drawdowns, ranges, volume, moving-average gaps	daily prices
Market	6	SPY trend and volatility, breadth, the regime	SPY, VIX
Fundamental	14	margins, returns on capital, growth, leverage, valuation (4), filing age	SEC company facts
Macro	8	CPI, unemployment, payrolls, financial conditions (as first published); curve, credit spread, Fed funds (lagged 1-2 days)	FRED / ALFRED vintages
Extra (zoo only)	17	6 SEC 8-K event counts, 8 analyst columns (firms, score, revisions, target upside, EPS surprise), 3 peer returns (10 most-correlated same-sector stocks)	SEC, Yahoo, prices

Cross-sectional features are ranked into [−0.5, 0.5] among that date's index members, so a model learns *which stocks* are better, not when the market is. Market and macro features stay as levels.

The clock: what a row predicts

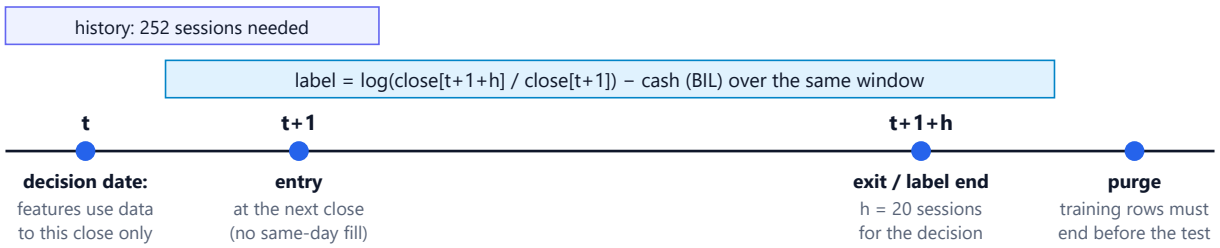


Figure 2. Features stop at the close of t; the trade enters at the next close; the label is the return from that entry over h sessions, above cash.

Costs

Each stock carries a round-trip cost: the Abdi-Ranaldo spread estimate over 21 sessions plus 2 basis points of slippage per side, clipped to 0.04%-5%. Half is charged at entry and half at exit. Every expected return the decision shows is **net** of it.

2. How the models are trained

Walk-forward testing

A model is only ever judged on years it did not see. For each test year from 2012, it is fitted on rows whose label ended before that year (the **purge**, so no training label overlaps the test period), then it scores the year. Skill is the average per-date Spearman rank correlation (IC) between the score and the realized net result, with a block-bootstrap 95% interval.

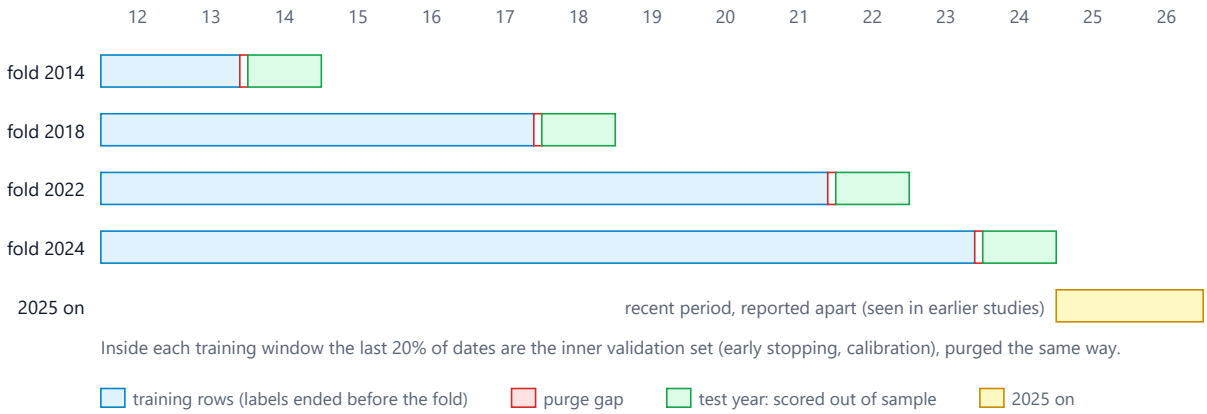


Figure 3. Expanding-window folds: train on everything that had ended, purge, score the next year. 2025 on is reported apart.

Refitting and reuse

Live models are fitted on *all* labelled history up to the last close, on decision dates every 5 sessions from 2006. A fit is kept and reused until it is 20 sessions old (Settings → schedule), then refitted. Before any work, a run computes an **input key**; when a saved run has the same key it is loaded, so the same inputs never produce a second, different output.

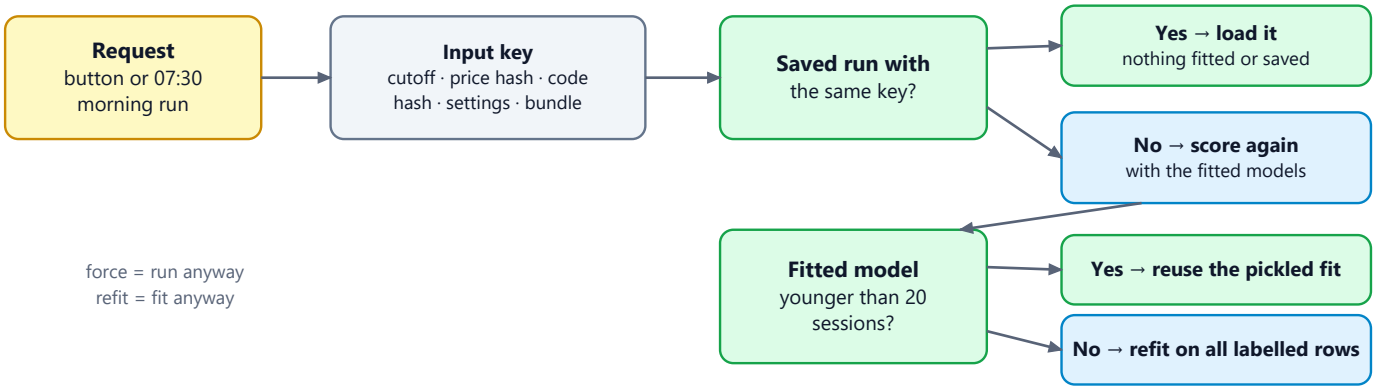


Figure 4. Same inputs → the saved output. New data → score with the cached fit, refitted every 20 sessions.

Run	Its input key covers	On the same key
Every model (zoo)	cutoff, that close's prices, watchlist, code of 6 files, ensemble file, Chronos pin	loads the saved run
Daily decision	cutoff, prices, rules, capital, ledger, watchlist, code, ensemble, zoo run, outcomes	returns the saved decision
Research forecast	tickers, model, horizon, bags, cutoff, prices, versions, code of 5 files	returns the saved run
Signals batch	tickers asked, universe, horizons, settings, cutoff, prices, code; the day if news is on	loads the saved run
Sector model	cutoff, prices, version, code	loads the saved forecast
AI agents	once a day per stock (the agents are not covered further here)	returns today's run

3. The model zoo

Every model ranks today's S&P 500 and the watchlist; the page shows each as a percentile (100 = its top pick). All learn the same label (20-session net log return over cash) unless noted. Hyperparameters below are fixed in code; nothing is tuned on the test years.

Model	Algorithm and settings	Inputs	Gives
XGB rank (54)	XGBoost pairwise ranking, grouped by date; relevance = within-date quintile; depth 3, eta 0.02, subsample 0.7, min child weight 200, λ 10; ≤ 2000 rounds, early stop 100 on the last 20% of dates, then refit on all	54 base	rank
XGB rank (all)	the same	71	rank
LGBM LambdaRank	LightGBM lambdarank (NDCG), 300 rounds, 15 leaves, lr 0.05, min 500 rows a leaf, feature/bagging 0.7, L2 10	71	rank
LGBM regression	LightGBM regression, 300 rounds, label clipped at 1%/99%	71	rank + return
Ridge pooled	Ridge α 1000 on median-filled, z-scored features	71	rank + return
Ridge per stock	one Ridge per stock (α 50), needs ≥ 80 of its own rows	33	rank + return
Mamba	state-space sequence model (Figure 5)	64-day sequence + 54	rank
Chronos-2	pretrained time-series model, zero-shot, 1024-day context; nothing is fitted	log closes	rank + return
Chronos-2 + market	the same with SPY, sector fund, volume, VIX, 10-year yield as covariates	prices	rank + return
Baselines	momentum 12-1, low volatility, analyst target upside, EPS surprise: one column each	1	rank

Roles: **decides** (the ensemble, else the decision's own XGB fit), **shown only** (every other model), **baseline** (simple yardsticks). One model failing never stops the others.

Mamba, the sequence model

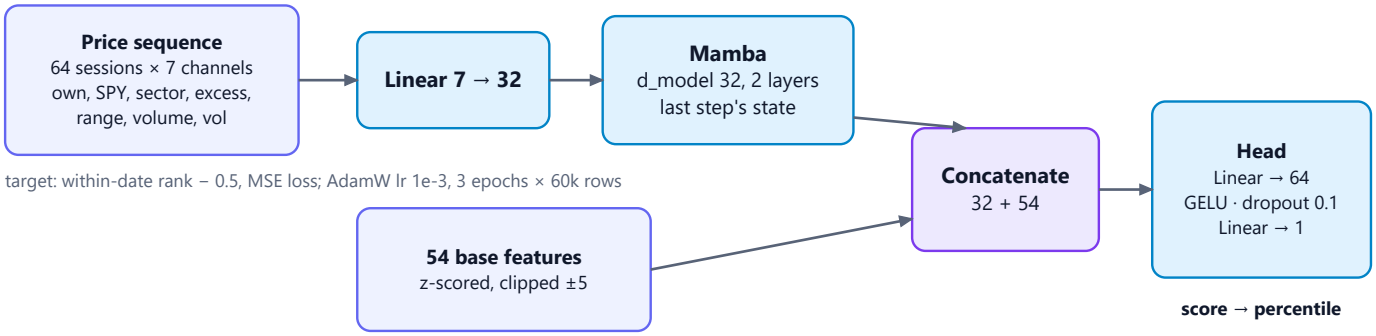


Figure 5. Mamba reads the last 64 sessions as a sequence, joins its summary to the static features and outputs a ranking score.

The 7 channels are the stock's log return, SPY's, the sector fund's, the stock minus SPY, the day's range, volume against its 20-day mean, and the log volatility; returns are divided by the stock's own volatility over the window. It runs on plain PyTorch (mambapy), CPU or GPU.

4. The ensemble: how the models are combined

Six zoo members, one per family, are combined by **stacking**: a second-stage fit that learns how much to trust each member from its *out-of-sample* record. The target is not the 20-session return but the rank of what the actual trade rule earns net of costs (the barrier trade of Figure 8), so the ensemble ranks stocks by the result the decision would get.

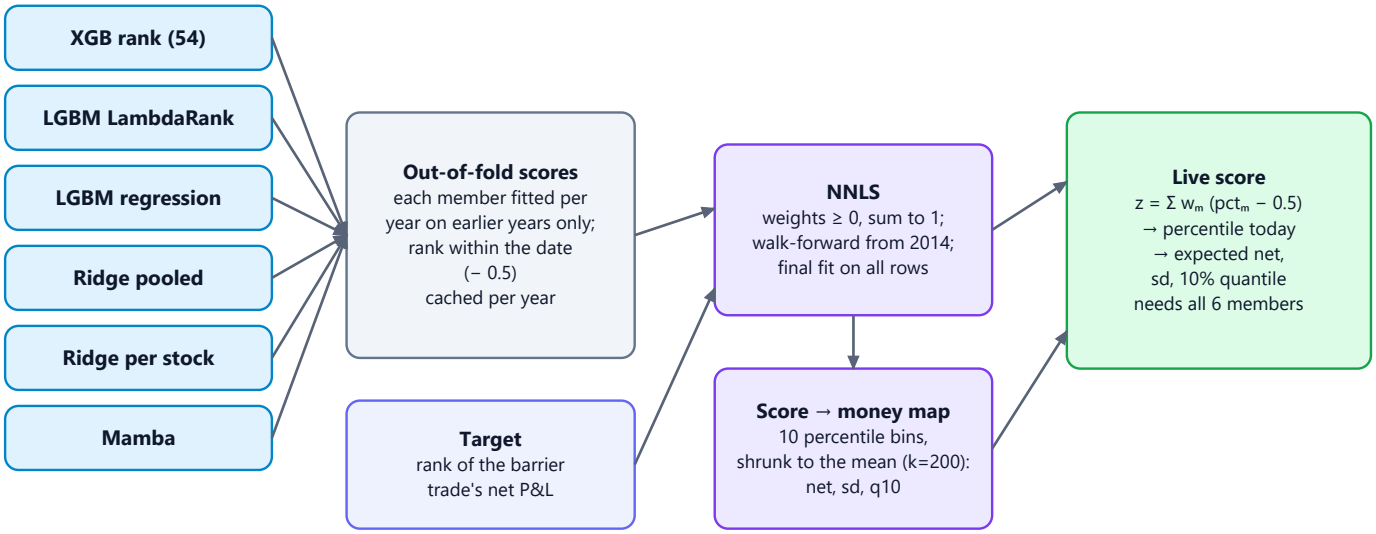


Figure 6. Out-of-fold member ranks → non-negative least squares → weights; the stacked percentile is mapped to money by 10 shrunk bins.

Learned weights

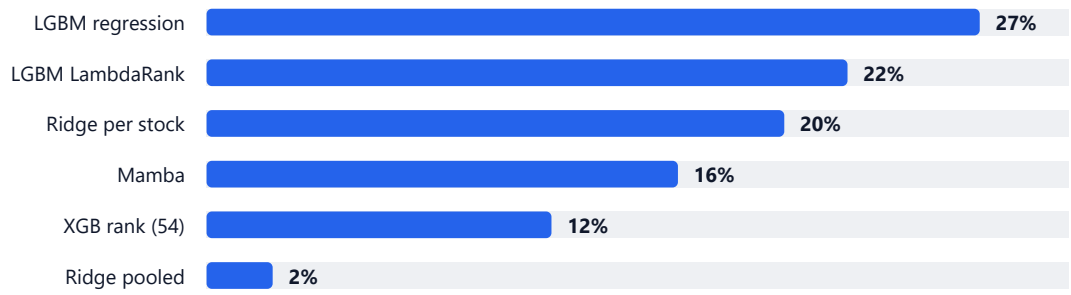


Figure 7. The current weights, learned on 285,697 rows through 2026-10-28. Non-negative and summing to 1, so no member is shorted.

Tested record

Score	IC 2014-24 (95% interval)	Top fifth − all	IC 2025 on (95% interval)
Stacked ensemble	0.032 (0.011, 0.053)	+0.25%	−0.010 (−0.055, 0.038)
Equal weights	0.043 (0.021, 0.063)	+0.39%	−0.003
LGBM regression alone	0.047		−0.016
Mamba alone	0.024		0.007
XGB rank alone	0.021		−0.001
LGBM LambdaRank alone	0.020		0.026
Ridge per stock alone	0.018		0.014
Ridge pooled alone	0.018		−0.017

The learned weights did **not** beat equal weights, and the 2025-on interval includes zero. Chronos-2 is reported beside it but given no weight (its training data may overlap the test years). Every stock needs all six members to get an ensemble score; the few without one (short histories for Ridge per stock) keep the decision's own XGB percentile.

5. The daily decision

Each morning after the close (07:30 by default), the decision scores every stock, applies the gates in order and sizes what passes. It considers the 25 best-ranked candidates and holds at most 5 names.

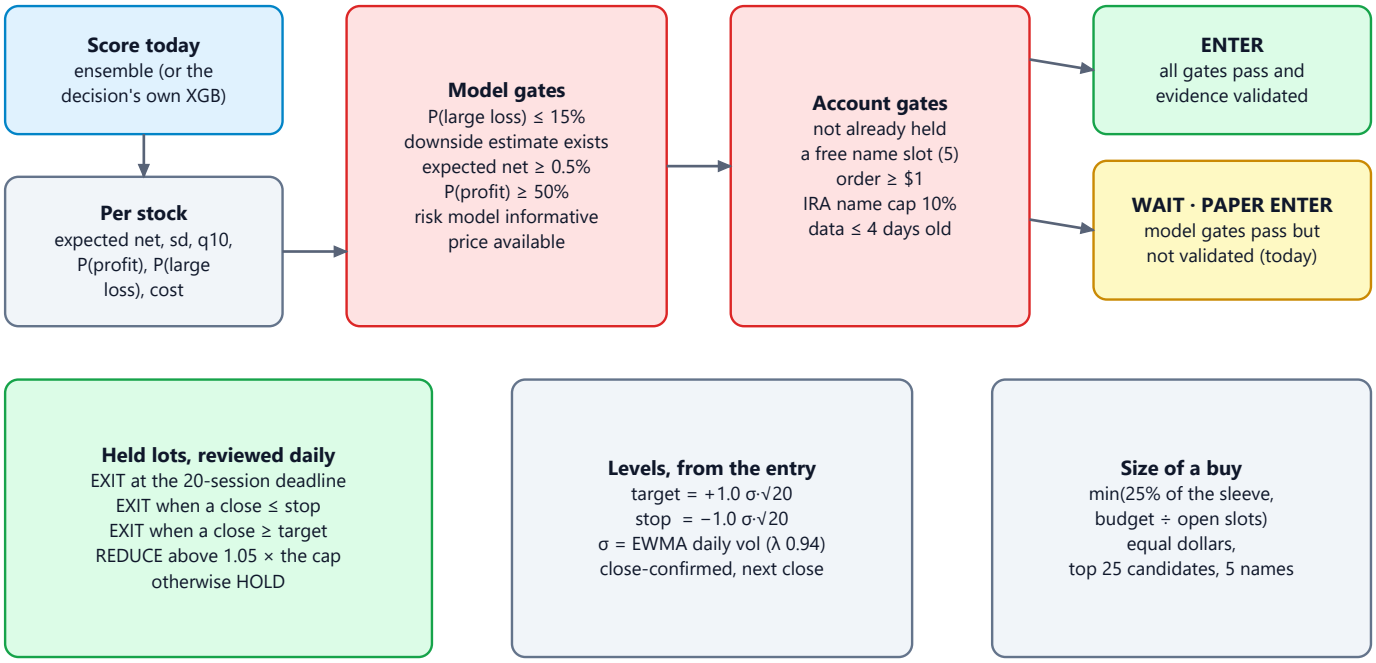


Figure 8. From score to action. Model gates judge the forecast; account gates judge the portfolio. Held lots are reviewed first.

The decision also fits its own models every 20 sessions: an XGBoost ranker (the fallback ranking), a calibrated logistic model for P(profit) and an XGBoost classifier for P(large loss), defined as the trade losing more than 1.5 σ over 20 sessions. A risk model that gives every stock nearly the same probability (spread under 0.5 points) counts as uninformative and blocks entries.

Validation rule. ENTER needs a matching study whose excess over SPY has a 95% interval above zero, beats cash, stays above a −35% drawdown, and a paper record of at least 20 round trips with positive excess. The saved study fails the SPY check (+210% against SPY's +358%), so today every pass is PAPER ENTER.

6. The signals (Screener)

A separate batch fits, for each horizon of 5, 20 and 60 sessions, the models below. It feeds the Screener's Signals view and the ticker page. Events are measured in each stock's own volatility units, σ_h = EWMA daily volatility × √h, so a calm and a wild stock are judged alike.

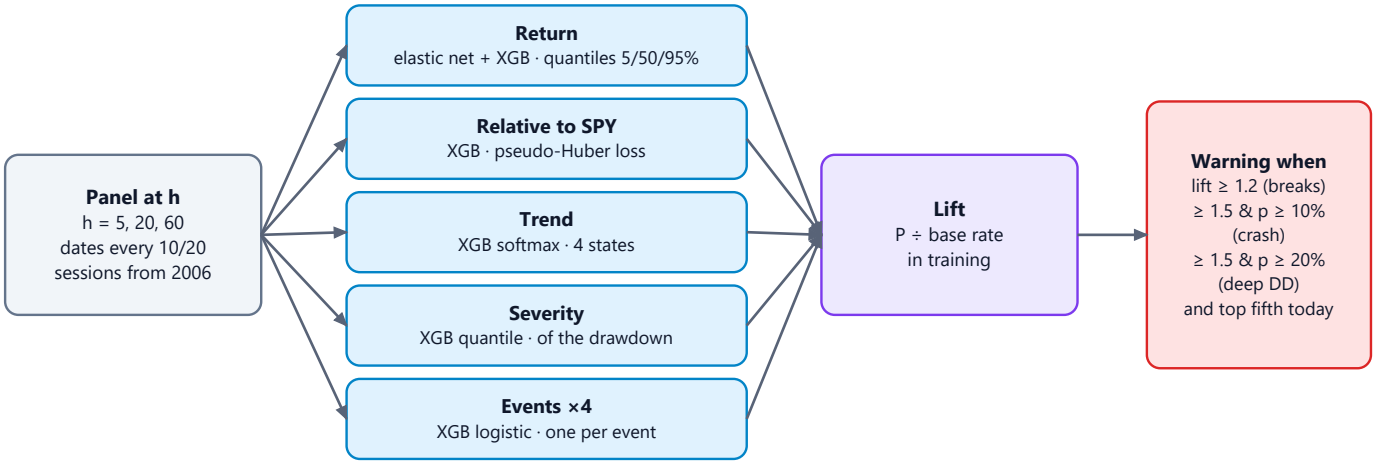


Figure 9. One panel per horizon, five model kinds, a lift against the training base rate, and warnings only where a stock also stands out today.

Event	Definition (over the next h sessions)
Breakout	any intraday high above the 20-day high
Breakdown	any intraday low below the 20-day low
Large loss (crash)	the h-session log return ≤ −1.5 σ_h
Deep drawdown	the lowest low ≤ −1.5 σ_h below today's close
Trend up / down	|return| ≥ 0.5 σ_h and path efficiency ≥ 0.30
Sideways	|return| ≤ 0.35 σ_h and range ≤ 1.25 σ_h; otherwise unclear

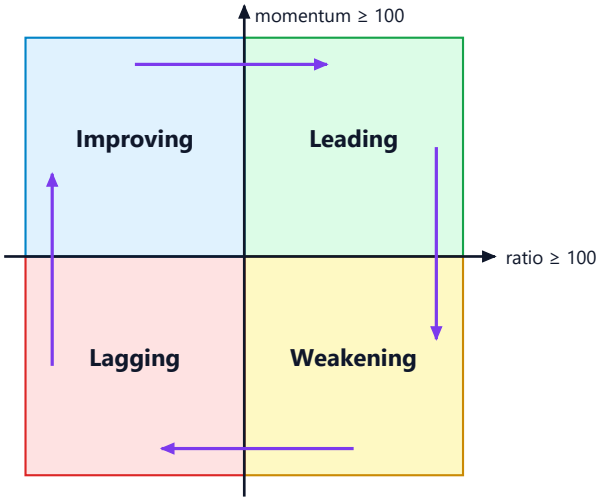
A warning also needs the stock in today's top fifth for that probability, and the model must spread its probabilities (10-90% range ≥ 2 points), so a market-wide shift does not flag everyone.

7. Research forecasts (ETFs)

For the 39 long-lived ETFs and your holdings, the research engine forecasts 5, 20 and 60 sessions ahead with simple and machine-learning models: zero, the training mean, a volatility prior, momentum, an elastic net (regularization chosen on the inner split), XGBoost (regression, ranking, quantiles) and Chronos-2. Prediction intervals come from the inner holdout's residual quantiles scaled by volatility.

Models are compared on yearly walk-forward blocks with Holm-adjusted p-values. A combination weights a model by its IC only if IC > 0 and Holm p < 0.05; Chronos-2 always gets 0. Saved ETF results at 20 sessions: XGBoost IC 0.113 (p 0.004), XGB quantile 0.107, elastic net 0.111. These predate the current code.

8. Sectors: rotation and the sector forecast



How it is measured

RS = sector ÷ SPY

ratio = 100 × RS ÷ its 63-day mean

momentum = 100 × ratio ÷ ratio 10 sessions ago

Alerts

Improving: early in

Weakening: early out

Leading or Lagging for ≤ 5 sessions

Why it moved

a ridge fit of the sector's excess return on 6 drivers: yields, curve, oil, dollar, credit, VIX

Figure 10. The rotation stages, from each sector's strength against SPY and its momentum. Sectors usually move clockwise.

The **sector forecast** ranks the 11 SPDR sector funds for the next 20 sessions with LightGBM LambdaRank (7 leaves, lr 0.03, 300 rounds) plus a regression for the size of the excess return over SPY. Its features are relative returns over 1 week to 1 year and their ranks, the rotation ratio and momentum, the 200-day gap, drawdown, relative volatility, and macro moves (yields, curve, oil, dollar, credit, VIX, SPY). Yearly folds 2010-24, holdout from 2025. Status: **not validated** (holdout IC 0.083, interval −0.038 to 0.183). The probable winner of a sector is its best-ranked stock in the daily decision.

9. The quality screen

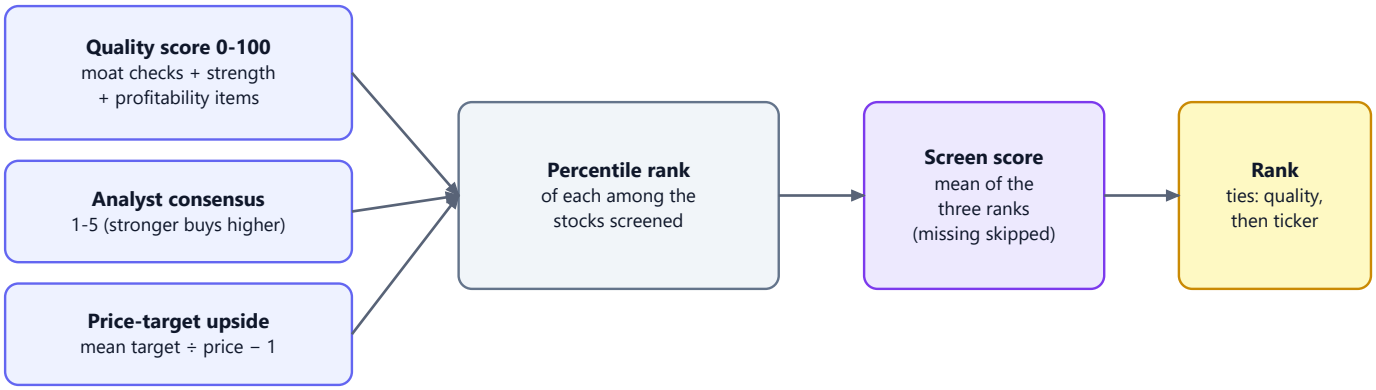


Figure 11. The Screener's score: three facts-and-opinion inputs, each turned into a percentile, averaged.

Moat (5 checks against sector bars): pricing power (gross margin level and stability), ROIC every year, operating margin, free cash flow positive every year, revenue growth in all but one year; banks use ROE, ROA and book-value growth. **Strength**: debt/equity, interest coverage, current ratio, a downturn test. **Valuation** is kept apart (reverse DCF at 9%). The screen is not a forecast and is untested for returns.

10. The feedback loop

Every stock the decision scores is saved *before* its outcome is known, with a procedure id that changes whenever the method (not the fitted weights) changes. After the trade's window, the outcome is replayed with the same execution rule and costs. Once a procedure has at least 300 resolved rows over 10 dates, its probabilities are recalibrated on its own live record.

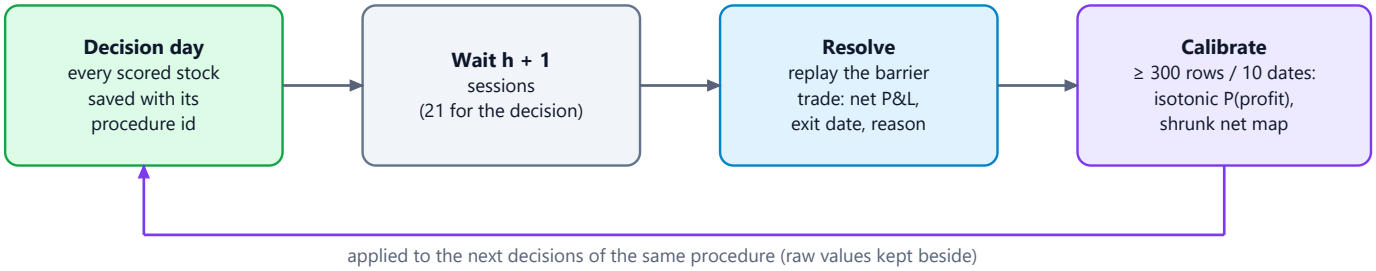


Figure 12. Predict, wait, resolve, calibrate: the live record checks the backtests.

11. What the evidence says

Model	IC 2012-24	IC 2025 on	Status
XGB rank (54)	0.022 (0.008, 0.037)	0.001	shown only
XGB rank (all 71)	0.032	−0.006	shown only
LGBM LambdaRank	0.020	0.021	ensemble member
LGBM regression	0.030	−0.003	ensemble member
Ridge pooled	0.015	−0.018	ensemble member
Ridge per stock	0.015	0.018	ensemble member
Mamba	0.018 (−0.003, 0.039)	0.006	ensemble member
Chronos-2 (zero-shot)	−0.006 (2023-24)	0.019 (0.011, 0.063)	shown, weight 0
Momentum 12-1	0.004	−0.013	baseline
Low volatility	0.002	−0.043	baseline
Analyst target upside	0.024	−0.007	baseline
EPS surprise	0.006	0.023	baseline

IC: average per-date rank correlation with the realized 20-session net result. 2012-24 has 654 dates; 2025 on has 86, so its intervals are wide. Adding the extra features to XGB raised its development IC from 0.022 to 0.032 (news 0.027, peers 0.026, consensus 0.021).

Known limits

- Every interval for 2025 on includes zero except Chronos-2 zero-shot, measured on only 20 dates.
- The 2025-on period was looked at in earlier studies, so it is not a clean holdout.
- P(profit) after the ensemble comes from a model fitted on the XGB percentile: a known mismatch.
- Stocks without an ensemble score keep the XGB percentile and are sorted together with ensemble percentiles.
- S&P 500 membership before the recorded history is today's list, so early years carry survivorship bias.
- The research forecasts' p-values assume independent dates, which overstates their confidence.

Generated from the code and saved studies on 29 October 2026.